Theme 3 Differentiation rules
Unit 3.1 The sum and constant rules

Note Title 2015/03/22

Notation:

$$\frac{d}{dx}\left(x^2\right) = 2x$$

$$y = x^2 \implies \frac{dy}{dx} = 2x$$

$$f(x) = x^2 \implies f'(x) = 2x.$$

Not okay: $\frac{d}{dx} = 2x$

$$f'(x) = 2x$$

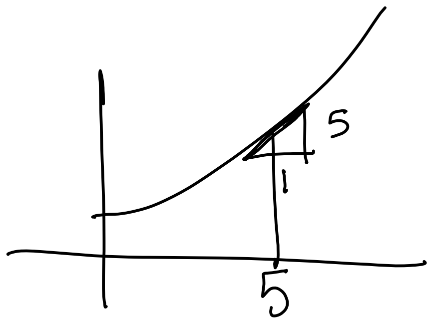
Higher order derivatives:

$$f(x) = x^3 \implies f'(x) = 3x^2 \implies f''(x) = 6x$$

$$\implies f'''(x) = 6 \implies f^{(4)}(x) = 0 \;\ldots\ldots \implies f^{(100)}(x) = 0.$$

Rules:

1. $\frac{d}{dx}\, c = 0$
2. $\frac{d}{dx}\, x^n = n\, x^{n-1}$
3. $\frac{d}{dx}\, e^x = e^x.$



Examples:

1. $\frac{d}{dx}\left(x^5\right) = 5x^4$
2. $\frac{d}{dx}\left(\frac{1}{x^2}\right) = -2x^{-3}$
3. $\frac{d}{dx}\sqrt{x} = \frac{1}{2}x^{-1/2}$
4. $\frac{d}{dx}(\pi x) = \pi$
5. $\frac{d}{dx}\, 6^3 = 0.$

$$\frac{d}{dx}\, x^n = n\, x^{n-1}$$

Rules:

1. Constant rule:

$$\frac{d}{dx}\left(c\, f(x)\right) = c\,\frac{d}{dx} f(x)$$

$$2 \cdot e^{x} = 2 \cdot e^{x}$$

2. Sum rule / Difference rule

$$\frac{d}{dx}\left(f(x) + g(x)\right) = \frac{d}{dx} f(x) + \frac{d}{dx} g(x).$$

$$\frac{d}{dx}\left(f(x) - g(x)\right) = \frac{d}{dx} f(x) - \frac{d}{dx} g(x).$$

Examples

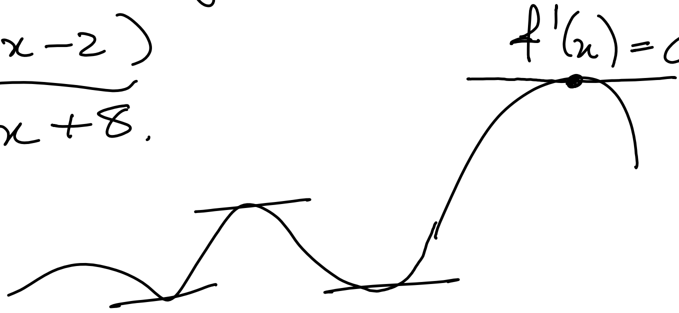
1. $$\frac{d}{dx}\left(5x^{1/3} - 2x^{5}\right) = \frac{5}{3}x^{-2/3} - 10x^{4}$$

2. $$\frac{d}{dx}\left(2e^{x} - \frac{4}{x^{2}}\right) = 2e^{x} + 8x^{-3}$$

3. $$\frac{d}{dx}\left(x^{1/2} - \frac{4}{x^{5}}\right) = \frac{1}{2}x^{-1/2} + \underbrace{20x^{-6}}_{\frac{20}{x^{6}}}.$$

More examples:

1. Find the the horizontal tangent line(s) to $f(x) = \underbrace{(x-4)(x-2)}_{x^{2}-6x+8.}$



$$f'(x) = 2x - 6 = 0$$
$$\Rightarrow x = 3.$$

Tangent line: $x = 3 \Rightarrow y = -1.$

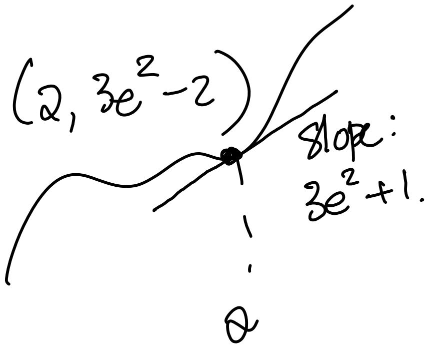
Horizontal tangent: $\underline{y = -1.}$

2. Find the equation of the tangent line to $f(x) = 3e^{x} - \frac{4}{x}$ at $x = 2$.

slope

$f'(x) = 3e^{x} + \frac{4}{x^2}$

$\Rightarrow f'(2) = 3e^{2} + 1.$



Point: $(2, 3e^{2}-2)$.

$\Rightarrow y - y_0 = m(x - x_0)$

$\Rightarrow y - (3e^{2} - 2) = (3e^{2}+1)(x-2)$

$\Rightarrow y = (3e^{2}+1)x - 2(3e^{2}+1) + 3e^{2} - 2$

$\Rightarrow y = (3e^{2}+1)x - 3e^{2} - 4$

Technique mastering Section I

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